OPPERMANN INTERVALS IN $\mathbb{F}_q[t]$: COUNTEREXAMPLES IN CHARACTERISTIC 2 AND A COMPUTER-ASSISTED THEOREM IN ODD CHARACTERISTIC

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ABSTRACT. For a monic $f \in \mathbb{F}_q[t]$ of degree d the Legendre interval $\{f^2 + s : \deg s \le d\}$ splits into the q *Oppermann sub-intervals* $O_{f,c} = \{f^2 + cf + s : \deg s < d\}$, $c \in \mathbb{F}_q$, each of size q^d; for $q = 2$ these are the two halves, the analogue of (n^2, n^2+n) and $(n^2+n, (n+1)^2)$. We ask whether every $O_{f,c}$ contains an irreducible polynomial. In characteristic 2 the answer is no, and the failures are explained completely by three Galois-theoretic obstructions: (A) for $d = 1$ the piece $c = 0$ consists of squares; (B) for $d = 2$ we compute the exact number of irreducibles in every Oppermann sub-interval, $q^2/4$, 0, $q(q-2)/4$ or $q(q-1)/2$, by letting Frobenius act on the roots of an additive polynomial through $\mathrm{AGL}_2(\mathbb{F}_2) \cong S_4$; it vanishes exactly when $q = 4^j$, $a_1^2 + c = 0$, or $q = 2$, $a_1^2 + c = 1$; (C) for $q = 2$, $d = 4$ the block $\{t^8 + s : \deg s \le 3\}$ contains no irreducible, by the Stickelberger–Swan sign of Frobenius, because $8 \mid \deg$ makes the sign depend only on the tail. In odd characteristic every class of the Hayes group of level d is an Oppermann sub-interval, and an exact computation with the Hayes explicit formula shows that all of them contain irreducibles for $q = 3$, $d \le 16$; $q = 5$, $d \le 10$; $q = 7$, $d \le 9$; $q = 9$, $d \le 8$; $q = 11$, $d \le 7$; $q = 13$, $d \le 6$. In characteristic 2 the same computation, for $q \in \{2, 4, 8, 16\}$ up to $q^d \le 2^{26}$, finds no empty Oppermann sub-interval beyond (A)–(C). The Weil bound sits at the square-root barrier for these intervals. For $d = 2$ it gives $\Psi \ge q$, and in the empty classes of (B) this is attained exactly, by fourth powers of linear polynomials. For fixed d and large q, the theorem of Bank, Bary-Soroker and Rosenzweig gives nonemptiness in most cases; we determine which.

1. INTRODUCTION

Oppermann's conjecture [6] asserts that for every $n \ge 2$ both (n^2, n^2+n) and $(n^2+n, (n+1)^2)$ contain a prime. It implies Legendre's conjecture and lies beyond the Riemann hypothesis, which gives primes only in intervals $[x, x + C\sqrt{x}\log x]$. Over $\mathbb{F}_q[t]$, Legendre intervals $I_f = \{f^2 + s : \deg s \le d\}$ (f monic of degree d) were studied in [3]. Since $f^2 + cf$ and $f^2 + ct^d$ differ by a polynomial of degree $< d$, the interval I_f is the disjoint union of the q sub-intervals

$$O_{f,c} = \{f^2 + cf + s : \deg s < d\} = \{f^2 + ct^d + s : \deg s < d\}, \qquad c \in \mathbb{F}_q, \tag{1}$$

each of cardinality $q^d = |f|$, the norm of f, exactly as (n^2, n^2+n) has length n. For $q = 2$ there are two halves. We call $O_{f,c}$ the *Oppermann sub-intervals* and ask:

Does every $O_{f,c}$ contain an irreducible polynomial?

The answer depends on the characteristic.

Theorem 1.1 (characteristic 2)**.** *Let $q = 2^k$.*

(A) *For $d = 1$ the sub-interval $O_{f,0} = \{t^2 + s_0 : s_0 \in \mathbb{F}_q\}$, which is the same set for all q polynomials f, consists of squares. Every $O_{f,c}$ with $c \ne 0$ contains exactly $q/2$ irreducibles.*

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(B) *For $d = 2$ and $f = t^2 + a_1 t + a_0$, the number of irreducibles in $O_{f,c}$ depends only on $e = a_1^2 + c$ and equals*

	$e = 0$	$e \neq 0$
k even	0	$q^2/4$
k odd	$q(q-1)/2$	$q(q-2)/4$

The table has two zero entries: k even with $e = 0$, and $q(q-2)/4$ at $q = 2$ (where $k = 1$ is odd). Hence $O_{f,c}$ is empty if and only if either k is even and $c = a_1^2$, or $q = 2$ and $c = a_1^2 + 1$. For $q = 2$ the block $e = 0$ is not *empty: it contains $q(q-1)/2 = 1$ irreducible, $t^4 + t + 1$. For every $q \in \{2, 4, 16, 64, \dots\}$ there is thus exactly one empty sub-interval as a set: B_0, or B_1 when $q = 2$. It is represented by q^2 pairs (f, c), one c for each of the q^2 polynomials f. For $q \in \{8, 32, 128, \dots\}$ there is none.*

(C) *For $q = 2$ and $d = 4$ the block $Z_4 = \{t^8 + s : \deg s \leq 3\}$ contains no irreducible polynomial. It equals $O_{f,c}$ for the 8 pairs $f = t^4 + a_2 t^2 + a_1 t + a_0$ (that is, $a_3 = 0$), $c = a_2$.*

Theorem 1.2 (odd characteristic, computer-assisted)**.** *Let q be odd. Every Oppermann sub-interval $O_{f,c}$ contains an irreducible polynomial whenever*

$$q = 3,\ d \leq 16; \qquad q = 5,\ d \leq 10; \qquad q = 7,\ d \leq 9;$$
$$q = 9,\ d \leq 8; \qquad q = 11,\ d \leq 7; \qquad q = 13,\ d \leq 6.$$

Equivalently (Lemma 2.1), every interval $\{P + s : \deg s < d\}$ with P monic of degree $2d$ contains an irreducible polynomial in these ranges.

Theorem 1.3 (characteristic 2, computer-assisted)**.** *For $q = 2$, $d \leq 26$; $q = 4$, $d \leq 13$; $q = 8$, $d \leq 7$; $q = 16$, $d \leq 5$, the only empty Oppermann sub-intervals are those of Theorem 1.1.*

For fixed d and large q, the theorem of Bank, Bary-Soroker and Rosenzweig [1] settles the question in most cases (Corollary 5.1). The tables above therefore matter mainly for small q, where that theorem gives nothing.

Throughout, a *sub-interval* means the set $O_{f,c}$. Distinct pairs (f, c) can give the same set; counts of sub-intervals are counts of sets, that is, of Hayes classes.

Section 2 identifies the Oppermann sub-intervals with Hayes classes and explains why characteristic 2 is special: the Oppermann classes are concentrated on prefixes close to squares. Section 3 proves Theorem 1.1. Section 4 describes the exact computation behind Theorems 1.2 and 1.3. Section 5 explains why the explicit formula alone cannot go further.

2. Oppermann sub-intervals as Hayes classes

For P monic of degree n write $P = t^n + p_{n-1}t^{n-1} + \cdots + p_0$. Two monic polynomials of degree n lie in the same *Hayes class of level l* if they agree in $p_{n-1}, \dots, p_{n-l}$. Each class is a coset $P + \{s : \deg s < n - l\}$, and the map

$$P \mapsto u^n P(1/u) \bmod u^{l+1} = 1 + p_{n-1}u + \cdots + p_{n-l}u^l$$

identifies the classes with the elements of the multiplicative group $G_l = (1 + u\mathbb{F}_q[u])/(1 + u^{l+1}\mathbb{F}_q[u])$ of order q^l. This map is multiplicative: $\mathrm{class}(PQ) = \mathrm{class}(P)\,\mathrm{class}(Q)$ in G_l. The characters of G_l are the short-interval characters of Hayes [4] and Keating–Rudnick [5]. A character has *exact level j* if it factors through G_j but not through G_{j-1}. There are $q^j - q^{j-1}$ characters of exact level j, for $1 \leq j \leq l$. By (1), $O_{f,c}$ is the Hayes class of level d and degree $n = 2d$ containing $f^2 + ct^d$.

Lemma 2.1 (odd q)**.** *If q is odd, every Hayes class of level d in degree $2d$ is an Oppermann sub-interval, and it equals $O_{f,c}$ for exactly q pairs (f, c).*

Proof. Write $f = \sum_{i=0}^{d} a_i t^i$ with $a_d = 1$. For $1 \leq j \leq d$ the coefficient of t^{2d-j} in f^2 is

$$[f^2]_{2d-j} = \sum_{\substack{i+i'=2d-j \\ 0 \leq i, i' \leq d}} a_i a_{i'} = 2a_{d-j} + \Phi_j, \qquad \Phi_j = \sum_{\substack{i+i'=2d-j \\ d-j < i,\, i' < d}} a_i a_{i'},$$

where the term $2a_{d-j} = 2a_d a_{d-j}$ comes from $\{i, i'\} = \{d, d-j\}$. The sum Φ_j runs over ordered pairs, so it includes the square $a^2_{d-j/2}$ when j is even. Φ_j depends only on $a_{d-1}, \dots, a_{d-j+1}$; for example $\Phi_1 = 0$, $\Phi_2 = a^2_{d-1}$, $\Phi_3 = 2a_{d-1}a_{d-2}$.

The map $(a_{d-1}, \dots, a_1) \mapsto ([f^2]_{2d-1}, \dots, [f^2]_{d+1})$ is therefore triangular with diagonal entries 2. Since $2 \in \mathbb{F}_q^\times$, it is a bijection $\mathbb{F}_q^{d-1} \to \mathbb{F}_q^{d-1}$: the prefix of a class determines $a_{d-1}, \dots, a_1$ uniquely, by solving $a_{d-j} = ([f^2]_{2d-j} - \Phi_j)/2$ for $j = 1, \dots, d-1$ in turn.

The remaining coefficient of the class is that of t^d in $f^2 + ct^d$, namely $2a_0 + \Phi_d + c$, where Φ_d is already fixed. For each of the q values of a_0 there is exactly one c giving the prescribed value. So every class arises, and from exactly q of the q^{d+1} pairs (f, c). □

So in odd characteristic the Oppermann question at degree $2d$ is the question whether *every* short interval of length $q^d = q^{n/2}$ contains a prime: no class is singled out.

Lemma 2.2 (q even)**.** *Let q be even. The Oppermann sub-intervals are exactly the Hayes classes of level d (in degree $2d$) in which the coefficient of t^j is 0 for every odd j with $d < j < 2d$. The coefficients of t^d and of t^j for even j with $d < j < 2d$ are arbitrary. There are $q^{\lceil d/2 \rceil}$ such classes among the q^d classes of level d, and each of them equals $O_{f,c}$ for exactly $q^{\lfloor d/2 \rfloor + 1}$ pairs (f, c).*

Proof. In characteristic 2, $f^2 = \sum a_i^2 t^{2i}$ has no odd terms, $a \mapsto a^2$ is bijective on $\mathbb{F}_q$, and the coefficient of t^d is shifted by c. This identifies the classes that occur.

For the count, note that $a \mapsto a^2$ is additive. So the map $(f, c) \mapsto \text{prefix}(f^2 + ct^d)$ is a surjective homomorphism of additive groups from $\mathbb{F}_q^{d+1}$, the coefficients $(a_0, \dots, a_{d-1}, c)$, onto $\mathbb{F}_q^{\lceil d/2 \rceil}$. Its fibres therefore all have size $q^{d+1-\lceil d/2 \rceil} = q^{\lfloor d/2 \rfloor + 1}$. □

For example, the empty classes of Theorem 1.1 at $d = 1$, 2 and 4 are represented by q, q^2 and $2^3 = 8$ pairs (f, c) respectively.

Thus in characteristic 2 the Oppermann sub-intervals are the short intervals around the $q^{\lceil d/2 \rceil}$ "near-squares" $f^2 + ct^d$, a set of measure $q^{-\lfloor d/2 \rfloor}$ among all short intervals. Squares and near-squares carry Galois-theoretic structure, and that structure produces the counterexamples below.

3. Characteristic 2: the three obstructions

Throughout this section $q = 2^k$ and $\mathrm{Tr} = \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}$.

3.1. **(A) $d = 1$.** Here $f = t + a_0$ and $O_{f,c} = \{t^2 + ct + s_0 : s_0 \in \mathbb{F}_q\}$.

For $c = 0$ the members $t^2 + s_0 = (t + \sqrt{s_0})^2$ are squares.

For $c \neq 0$, substitute $t = cy$: $t^2 + ct + s_0 = c^2(y^2 + y + s_0/c^2)$. This is irreducible if and only if $y^2 + y + s_0/c^2$ has no root in $\mathbb{F}_q$, that is, if and only if $\mathrm{Tr}(s_0/c^2) = 1$. Exactly $q/2$ values of s_0 satisfy this.

3.2. **(B) $d = 2$: an additive polynomial and $\mathrm{AGL}_2(\mathbb{F}_2)$.** For $f = t^2 + a_1 t + a_0$ we have $f^2 = t^4 + a_1^2 t^2 + a_0^2$, so

$$O_{f,c} = B_e := \{t^4 + et^2 + \alpha t + \beta : \ \alpha, \beta \in \mathbb{F}_q\}, \qquad e = a_1^2 + c.$$

By Lemma 2.2 the blocks B_e, $e \in \mathbb{F}_q$, are precisely the Oppermann classes for $d = 2$.

Proposition 3.1. *Let $\alpha \neq 0$. Put $L(t) = t^4 + et^2 + \alpha t$ and $R(y) = y^3 + ey + \alpha$.*

1. *$t^4 + et^2 + \alpha t + \beta$ is irreducible for some β if and only if R has exactly one root in $\mathbb{F}_q$.*
2. *If R has exactly one root in $\mathbb{F}_q$, then $t^4 + et^2 + \alpha t + \beta$ is irreducible for exactly $q/2$ values of $\beta \in \mathbb{F}_q$.*

If $\alpha = 0$, every member $t^4 + et^2 + \beta = (t^2 + \sqrt{e}\, t + \sqrt{\beta})^2$ is a square.

Proof. L is $\mathbb{F}_2$-linear and separable ($L' = \alpha \neq 0$), so $V = \ker L \subset \overline{\mathbb{F}}_q$ is a 2-dimensional $\mathbb{F}_2$-space, and $V \setminus \{0\}$ is the set of roots of $L(t)/t = R(t)$. The Frobenius $\varphi(x) = x^q$ acts on V by some $F \in \mathrm{GL}(V) \cong S_3$. Exactly one of three cases holds: $F = 1$ (R has three roots in $\mathbb{F}_q$), F of order 3 (no root), or F a transvection (exactly one root).

Fix β and a root r of $P_\beta = L + \beta$. The roots of P_β form the coset $r + V$. Since $L(\varphi(r)) = \varphi(\beta) = \beta$, we have $\varphi(r) = r + v$ with $v \in V$, and $\varphi(r + w) = r + (Fw + v)$. So Frobenius acts on the roots through the affine map $T(w) = Fw + v$ of $V \cong \mathbb{F}_2^2$, an element of $\mathrm{AGL}_2(\mathbb{F}_2) \cong S_4$. P_β is irreducible if and only if T is a 4-cycle.

- If $F = 1$, T is a translation and $T^2 = 1$.
- If F has order 3, $F - 1$ is invertible, so T has a fixed point.
- If F is a transvection, $T^2(w) = w + (F+1)v$, so T is a 4-cycle if and only if $(F+1)v \neq 0$, i.e. $v \notin V^F$.

This proves the "only if" part of (1); the "if" part follows from (2), since $q/2 \geq 1$. For (2), let F be a transvection, so $(F+1)V = V^F$ is a line. Replacing r by $r + w$ replaces v by $v + (F+1)w$, so the class $[v] \in V/V^F \cong \mathbb{F}_2$ depends only on β, and $\beta \mapsto [v]$ is additive. Its kernel consists of the β for which some root $r' = r + w$ is fixed by φ, i.e. $\beta \in L(\mathbb{F}_q)$. The map $L : \mathbb{F}_q \to \mathbb{F}_q$ is $\mathbb{F}_2$-linear with kernel $V^F = V \cap \mathbb{F}_q$ of order 2, so $|L(\mathbb{F}_q)| = q/2$ and exactly $q/2$ values of β have $[v] \neq 0$. □

Lemma 3.2. *Let $\alpha \neq 0$. The cubic $R(y) = y^3 + ey + \alpha$ has exactly one root in $\mathbb{F}_q$ if and only if $\mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)$.*

Proof. The roots y_1, y_2, y_3 of R are distinct (R is separable as above), with $y_1 + y_2 + y_3 = 0$ and $y_1 y_2 y_3 = \alpha$. Put $\rho = y_1/y_3 + y_2/y_1 + y_3/y_2$ and $\rho' = y_2/y_3 + y_1/y_2 + y_3/y_1$. Even permutations of the roots fix ρ and odd ones exchange ρ, ρ'; using $y_i + y_j = y_k$,

$$\rho + \rho' = \frac{y_1 + y_2}{y_3} + \frac{y_2 + y_3}{y_1} + \frac{y_3 + y_1}{y_2} = 1.$$

Hence ρ, ρ' are the roots of $z^2 + z + \delta$ with $\delta = \rho\rho' \in \mathbb{F}_q$, and Frobenius acts on the roots of R by an odd permutation (a transposition: exactly one root in $\mathbb{F}_q$) if and only if $\rho \notin \mathbb{F}_q$, i.e. $\mathrm{Tr}(\delta) = 1$. Expanding,

$$\delta = 3 + \sum_k \frac{y_i y_j}{y_k^2} + \sum_k \frac{y_k^2}{y_i y_j} = 1 + \sum_k \frac{\alpha}{y_k^3} + \frac{1}{\alpha} \sum_k y_k^3.$$

From $y_k^3 = ey_k + \alpha$ we get $\sum_k y_k^3 = \alpha$, so $\delta = \sum_k \alpha/w_k$ with $w_k = y_k^3 = ey_k + \alpha$. Substituting $y = (w + \alpha)/e$ (for $e \neq 0$) into $w = y^3$ shows that the w_k are the roots of $w^3 + \alpha w^2 + (\alpha^2 + e^3)w + \alpha^3$, so $\sum_k 1/w_k = (\alpha^2 + e^3)/\alpha^3$ and

$$\delta = 1 + e^3/\alpha^2.$$

For $e = 0$ this also holds directly ($w_k = \alpha$). So $\mathrm{Tr}(\delta) = 1$ if and only if $\mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)$. □

Remark 3.3. δ is Berlekamp's characteristic-2 discriminant [2] of R, and R is the cubic resolvent of the quartic; the proof above is self-contained. For $e = 0$ the lemma says: $y^3 = \alpha$ has exactly one root iff k is odd, which is the familiar fact that cubing is bijective on $\mathbb{F}_q$ iff $3 \nmid q - 1$.

Proof of Theorem 1.1(B). By Proposition 3.1, $|B_e \cap \{\text{irreducibles}\}| = \frac{q}{2} \cdot \#\{\alpha \neq 0 : \mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)\}$. If $e = 0$ the condition reads $\mathrm{Tr}(1) = 1$, i.e. k odd: the count is $\frac{q}{2}(q-1)$ for k odd and 0 for k even. If $e \neq 0$, $\alpha \mapsto e^3/\alpha^2$ permutes $\mathbb{F}_q^\times$, and $\#\{x \neq 0 : \mathrm{Tr}(x) = 1\} = q/2$, $\#\{x \neq 0 : \mathrm{Tr}(x) = 0\} = q/2 - 1$; with $\mathrm{Tr}(1) = k \bmod 2$ this gives $q^2/4$ (k even) and $\frac{q}{2}(\frac{q}{2} - 1)$ (k odd). The count $q(q-2)/4$ vanishes only for $q = 2$. □

The formula was checked by brute force for $q \leq 64$ (script `char2_d2_theorem.py`): criterion of Lemma 3.2, the $q/2$ count of Proposition 3.1(2), and the four block totals.

3.3. **(C) $q = 2$, $d = 4$: the sign of Frobenius.** The Oppermann classes for $q = 2$, $d = 4$ are the four prefixes $(p_7, p_6, p_5, p_4) = (0, x, 0, y)$ (Lemma 2.2). For $x = y = 0$ the class is the *zero block*

$$Z_4 = \{t^8 + a_3t^3 + a_2t^2 + a_1t + a_0\} \subset \mathbb{F}_2[t],$$

equal to $O_{f,c}$ for $f = t^4 + a_2t^2 + a_1t + a_0$ and $c = a_2$. Here $a_3 = 0$ is forced, because the coefficient of t^6 in f^2 is a_3^2; this is why there are $2^3 = 8$ pairs rather than 16.

We use the theorem of Stickelberger [7] in the form of Swan [8, Cor. 1]: *if $\tilde{P} \in \mathbb{Z}[t]$ is monic of degree n and its reduction $P \in \mathbb{F}_2[t]$ is squarefree with r irreducible factors, then $r \equiv n \pmod 2$ if and only if* $\operatorname{disc}(\tilde{P}) \equiv 1 \pmod 8$. Equivalently, the Frobenius permutation of the roots of P is even if and only if $\operatorname{disc}(\tilde{P}) \equiv 1 \pmod 8$. An irreducible P of degree 8 has Frobenius an 8-cycle, an odd permutation, and would need $\operatorname{disc}(\tilde{P}) \equiv 5 \pmod 8$.

Lemma 3.4. *Let $\tilde{P} = t^n + \tilde{g}$ with $\tilde{g} \in \mathbb{Z}[t]$, $\deg \tilde{g} < n$, and $8 \mid n$. Then* $\operatorname{disc}(\tilde{P}) \equiv \operatorname{Res}(\tilde{P}, \tilde{g}') \pmod 8$.

Proof. $\operatorname{disc}(\tilde{P}) = (-1)^{n(n-1)/2} \operatorname{Res}(\tilde{P}, \tilde{P}') = \prod_{\tilde{P}(\rho)=0}(n\rho^{n-1} + \tilde{g}'(\rho))$, the sign being $+1$ for $8 \mid n$. Expanding the product, every term containing a factor n is n times a symmetric polynomial with integer coefficients in the roots, hence lies in $8\mathbb{Z}$. □

Proof of Theorem 1.1(C). An irreducible member of Z_4 has no root in $\mathbb{F}_2$, so $a_0 = 1$ and $a_1 + a_2 + a_3 = 1$. This leaves four polynomials. $t^8 + t^2 + 1 = (t^4 + t + 1)^2$ is a square. For the other three choose integer lifts and apply Lemma 3.4:

- $t^8 + t + 1$: $\tilde{g}' = 1$, $\operatorname{disc} \equiv 1$;
- $t^8 + t^3 + 1$: $\tilde{g}' = 3t^2$, $\operatorname{Res}(\tilde{P}, 3t^2) = 3^8\big(\prod \rho\big)^2 = 3^8 \equiv 1$;
- $t^8 + t^3 + t^2 + t + 1$, lifted as $t^8 + (t + 1)^3$: $\tilde{g}' = 3(t + 1)^2$, and $\operatorname{Res}(\tilde{P}, 3(t + 1)^2) = 3^8 \prod(\rho + 1)^2 = 3^8\tilde{P}(-1)^2$, with $\tilde{P}(-1) = (-1)^8 + 0^3 = 1$, so $\operatorname{disc} \equiv 3^8 \equiv 1$.

Suppose one of these three were irreducible over $\mathbb{F}_2$. Then it would be squarefree with $r = 1$, and Stickelberger–Swan would require $\operatorname{disc}(\tilde{P}) \equiv 5 \pmod 8$. That contradicts $\operatorname{disc}(\tilde{P}) \equiv 1$, so none of them is irreducible. The argument does not need squarefreeness to be checked beforehand. □

Remark 3.5. The mechanism is that $8 \mid \deg P$ kills the leading term of P' modulo 8, so the sign of Frobenius is decided by the tail g alone; with $\deg g \le 3$ the tail is too short to produce the sign -1 on a polynomial with no linear factor. Table 1 shows the zero blocks $Z_d = \{t^{2d} + s : \deg s < d\}$, which are Oppermann classes for every d. For $d = 8$ ($8 \mid 2d$ again) the tail has degree ≤ 7, half of the squarefree members already have the sign -1, and the block contains 10 irreducibles. The parallel with (B) is exact: in (B) the obstruction is the quotient $S_4 \to S_3$ (the resolvent cubic), in (C) the quotient $S_8 \to \{\pm 1\}$ (the sign).

4. The exact computation

4.1. **Method.** Here Λ is the von Mangoldt function: $\Lambda(F) = \deg \varpi$ if $F = \varpi^k$ for a monic irreducible ϖ and some $k \ge 1$, and $\Lambda(F) = 0$ otherwise. For a class x of level l and degree n, let $\Psi(x) = \sum_{F \in x} \Lambda(F)$. The explicit formula [4, 5] reads

$$\Psi(x) = q^{n-l} + q^{-l} \sum_{\chi \neq 1} \overline{\chi}(x)\, \psi_n(\chi), \qquad u\frac{L'(u,\chi)}{L(u,\chi)} = \sum_{m \ge 1} \psi_m(\chi) u^m, \tag{2}$$

where for $\chi \neq 1$ the L-function $L(u, \chi) = \sum_F \chi(F) u^{\deg F}$ is a polynomial of degree $\le l - 1$. Its coefficients are the Fourier transforms on G_l of the class distributions of the monic polynomials of degree $< l$; $\psi_n(\chi)$ follows by the Newton recursion, and one inverse FFT on G_l returns $\Psi(x)$ for all q^l classes at once. This is the algorithm of [3], run here at level $l = d$ and degree $n = 2d$. The number of irreducibles in x is

$$\pi(x) = \frac{1}{n}\big(\Psi(x) - E(x)\big), \qquad E(x) = \sum_{k \ge 2,\ k \mid n} \ \sum_{\deg \varpi = n/k,\ \varpi^k \in x} \frac{n}{k}.$$

TABLE 1. The zero blocks $Z_d = \{t^{2d} + s : \deg s < d\} \subset \mathbb{F}_2[t]$. Columns: the squarefree members; those with disc $\equiv 5 \pmod 8$, i.e. Frobenius odd (integer lift with coefficients $0, 1$); those among them with no root in $\mathbb{F}_2$; and the irreducibles. For $d = 4$ the two members with disc $\equiv 5$ are t^8+t^3+t and $t^8+t^3+t^2+1$, which have the roots 0 and 1. Stickelberger–Swan holds on every squarefree member (script `swan_parity.py`).

d	$\lvert Z_d\rvert$	squarefree	disc $\equiv 5$	disc $\equiv 5$, no root	irreducible
2	4	2	2	1	1
3	8	4	3	1	1
4	16	8	2	**0**	**0**
5	32	16	10	4	4
6	64	32	20	8	6
7	128	64	42	17	9
8	256	128	64	24	10
9	512	256	130	61	28
10	1024	512	280	120	49

The proper prime powers in $E(x)$ are enumerated directly (Ben-Or test in degree $\le d$) and placed by the rule $\mathrm{class}(\varpi^k) = \mathrm{class}(\varpi)^k$ in G_l. Each run checks that every $\pi(x)$ is an integer and that $\sum_x (n\pi(x) + E(x)) = q^n$, and reports the largest distance of the floating-point $\Psi(x)$ from an integer (below 10^{-7} in all runs). As an independent check, a brute-force census that tests every polynomial of degree $2d$ for irreducibility (`verify_pointwise_oppermann.py`) agrees with the FFT counts class by class for all $q^{2d} \le 2^{22}$ (`compare_bruteforce.py`), and in minimum, mean and empty classes for all $q^{2d} \le 2^{26}$.

4.2. **Results.** Table 2 lists, for odd q, the minimum of $\pi(x)$ over all classes (all of which are Oppermann sub-intervals by Lemma 2.1) and its ratio to the expected value $q^d/2d$. Table 3 gives the same for the Oppermann classes in characteristic 2. The full output, all degrees, is in the file
`data/hayes_oppermann_census.txt`.

Proof of Theorems 1.2 and 1.3. Exact computation as described above; the minima are in Tables 2 and 3, and the empty classes in characteristic 2 occur exactly at

$$(q, d) \in \{(2,1), (2,2), (2,4), (4,1), (4,2), (8,1), (16,1), (16,2)\},$$

one class each, which are the classes of Theorem 1.1. □

The normalised minimum $\min_x \pi(x)/(q^d/2d)$ tends to 1 in every column: the worst Oppermann sub-interval is only slightly below average, far from the Weil bound discussed next.

TABLE 2. Odd q: irreducible counts π over all q^d Hayes classes of level d in degree $2d$ (all are Oppermann sub-intervals). Last four degrees for each q; all smaller degrees are in the data file, with minimum ≥ 1 throughout.

q	d	classes	$\min \pi$	mean π	$\min \pi/(q^d/2d)$
3	13	1 594 323	60 556	61 320.1	0.9875
	14	4 782 969	169 300	170 820.3	0.9911
	15	14 348 907	476 000	478 296.9	0.9952
	16	43 046 721	1 341 030	1 345 210.0	0.9969
5	7	78 125	5 418	5 580.3	0.9709
	8	390 625	23 951	24 414.0	0.9810
	9	1 953 125	107 410	108 506.9	0.9899
	10	9 765 625	486 042	488 281.2	0.9954
7	6	117 649	9 500	9 804.0	0.9690
	7	823 543	57 981	58 824.4	0.9857
	8	5 764 801	358 343	360 300.0	0.9946
	9	40 353 607	2 235 915	2 241 867.0	0.9973
9	5	59 049	5 680	5 904.8	0.9619
	6	531 441	43 728	44 286.7	0.9874
	7	4 782 969	339 993	341 640.6	0.9952
	8	43 046 721	2 684 630	2 690 420.0	0.9978
11	4	14 641	1 735	1 830.0	0.9480
	5	161 051	15 804	16 105.0	0.9813
	6	1 771 561	146 403	147 630.0	0.9917
	7	19 487 171	1 388 036	1 391 940.7	0.9972
13	3	2 197	349	366.0	0.9531
	4	28 561	3 450	3 570.0	0.9664
	5	371 293	36 672	37 129.2	0.9877
	6	4 826 809	400 104	402 234.0	0.9947

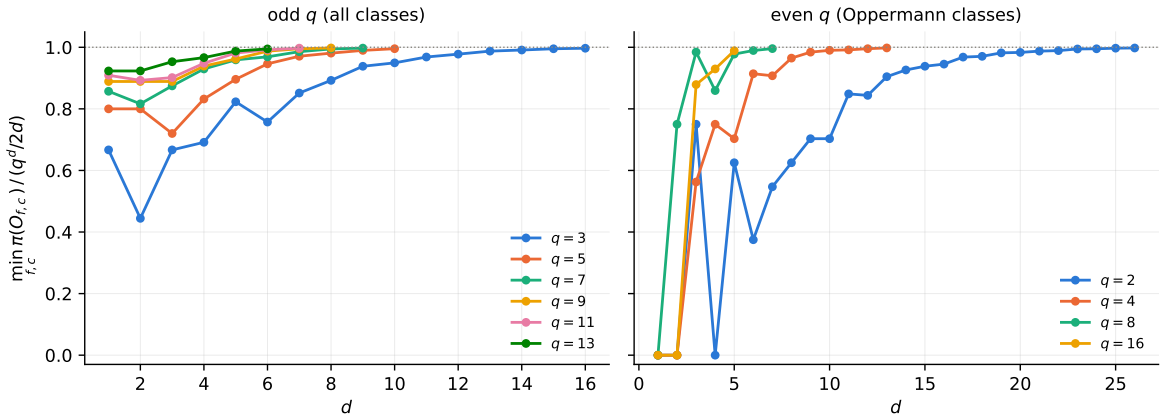


FIGURE 1. Normalised minimum $\min_{f,c} \pi(O_{f,c})/(q^d/2d)$ against d. Left: odd q. Right: even q; the points at 0 are the empty sub-intervals of Theorem 1.1.

TABLE 3. Even q: irreducible counts over the $q^{\lceil d/2 \rceil}$ Oppermann classes (Lemma 2.2). Last four degrees for each q; the only empty classes in the whole range are those of Theorem 1.1.

q	d	classes	$\min \pi$	mean π	$\min \pi/(q^d/2d)$
2	23	4 096	181 349	182 329.3	0.9945
	24	4 096	347 822	349 412.9	0.9951
	25	8 192	669 169	671 025.2	0.9971
	26	8 192	1 287 576	1 290 408.9	0.9977
4	10	1 024	51 908	52 350.0	0.9901
	11	4 096	189 054	190 603.5	0.9916
	12	4 096	695 868	698 820.5	0.9954
	13	16 384	2 575 566	2 580 963.0	0.9979
8	4	64	440	490.0	0.8594
	5	512	3 204	3 276.0	0.9778
	6	512	21 616	21 840.0	0.9895
	7	4 096	149 148	149 796.0	0.9957
16	2	16	**0**	60.0	**0**
	3	256	600	680.0	0.8789
	4	256	7 616	8 160.0	0.9297
	5	4 096	103 680	104 856.0	0.9888

5. The square-root barrier

In (2) with $n = 2d$ and $l = d$ the main term is q^d.

Let χ have exact level j. The monic polynomials of degree $\geq j$ are equidistributed over G_j, so $\deg L(u,\chi) \leq j-1$. By Weil, every inverse zero γ of $L(u,\chi)$ has $|\gamma| \in \{1, \sqrt{q}\}$, hence $|\psi_{2d}(\chi)| \leq (j-1)q^d$. Characters of level 1 have $L = 1$ and contribute nothing. Multiplying by the factor q^{-d} in (2), which cancels the q^d, and summing over the $q^j - q^{j-1}$ characters of each exact level j gives

$$|\Psi(x) - q^d| \leq \sum_{j=2}^{d} (j-1)(q^j - q^{j-1}) = (d-1)q^d - \sum_{j=1}^{d-1} q^j. \tag{3}$$

For $d \geq 3$ the right-hand side exceeds q^d for every q, so (3) does not even give $\Psi(x) > 0$. This is the exact analogue of the situation over $\mathbb{Z}$: Oppermann's intervals have length $\sqrt{x}$, and the Riemann hypothesis falls short by a factor $\log x$. No degree threshold beyond which $O_{f,c} \neq \emptyset$ follows from Weil's theorem.

For $d = 2$, (3) reads $|\Psi(x) - q^2| \leq q^2 - q$, so $\Psi(x) \geq q > 0$ for every class. This still does not produce an irreducible, because Ψ also counts prime powers. Theorem 1.1(B) shows the lower bound is sharp. Take $q = 4^j$ and the empty class $B_0 = \{t^4 + \alpha t + \beta\}$. Its proper prime powers are the q fourth powers $(t + c_0)^4 = t^4 + c_0^4$, each with $\Lambda = 1$; the squares $(t^2 + b)^2$ do not count, since $t^2 + b$ is itself a square. Hence

$$\Psi(B_0) = 4\pi(B_0) + E(B_0) = 0 + q = q.$$

So equality holds in (3), which forces $\overline{\chi}(B_0)\psi_4(\chi) = -q^2$ for all $q^2 - q$ characters of exact level 2. The whole of the Weil lower bound is taken up by the fourth powers.

The same happens for $q = 2$ on B_1, where $\Psi(B_1) = E(B_1) = 2$ comes from $(t^2+t+1)^2$. The values $\Psi(B_0) = q$ for $q = 4, 16$ and $\Psi(B_1) = 2$ for $q = 2$ were confirmed with the FFT code.

Results on the variance of Ψ over Legendre intervals (an average over f) bound the number of intervals that deviate by a given amount; they do not exclude a single empty interval.

For fixed degree and $q \to \infty$, Bank, Bary-Soroker and Rosenzweig [1] prove the following. For monic $P \in \mathbb{F}_q[t]$ of degree k and $1 \le m < k$, the number N of irreducibles in $\{P+h : \deg h \le m\}$ satisfies

$$|N - q^{m+1}/k| \le c(k)\, q^{m+1/2},$$

provided that $m \ge 2$ if $\gcd(q, k(k-1)) > 1$, and $m \ge 3$ if q is even and $\deg P' \le 1$. They also note that the estimate fails outside this range.

Our blocks $B_e = \{t^4 + et^2 + h : \deg h \le 1\}$ have $k = 4$ and $m = 1$, and in characteristic 2 the centre $P = t^4 + et^2$ has $P' = 0$, so both conditions exclude them. Theorem 1.1(B) gives the exact count at this excluded point and shows how the asymptotic fails:

- for $q = 4^j$ the block $e = 0$ contains no irreducible, while the expected count is $q^2/4$;
- for $q = 2^{2j+1}$ the block $e = 0$ contains $q(q-1)/2$, about twice the expected count.

The blocks $e \neq 0$, with $q^2/4$ and $q(q-2)/4$ irreducibles, do agree with the asymptotic. The obstruction comes from the additive structure of $t^4 + et^2 + \alpha t$ (Proposition 3.1). For $m = 1$, $k = 4$ and q even, the conditions of [1] therefore cannot be dropped.

Theorem 1.1(C) has $q = 2$ fixed, so it lies outside the $q \to \infty$ regime altogether.

In the other direction, [1] gives nonemptiness for large q in most cases. An Oppermann sub-interval $O_{f,c} = \{f^2 + ct^d + h : \deg h \le d-1\}$ has $k = 2d$ and $m = d-1$. Since $N \ge q^d/2d - c(2d)q^{d-1/2}$, the sub-interval is nonempty as soon as $q > (2d\, c(2d))^2$, provided the conditions of [1] hold. We do not make the constant $c(2d)$ explicit.

Corollary 5.1 (of [1])**.** *For each d there is $q_0(d)$ such that every Oppermann sub-interval of degree $2d$ over $\mathbb{F}_q$, with $q \ge q_0(d)$, contains an irreducible polynomial in each of the following cases:*

1. *q odd and $d \ge 3$;*
2. *q odd, $d = 2$ and $\gcd(q, 6) = 1$;*
3. *q even and $d \ge 4$;*
4. *q even, $d = 3$, and the class has $p_3 \neq 0$.*

Proof. We check the conditions of [1] with $k = 2d$ and $m = d-1$; recall that $1 \le m < k$ is required.

- *q odd.* The only condition is $m \ge 2$ when $p \mid 2d(2d-1)$. It holds for $d \ge 3$. For $d = 2$, $k(k-1) = 12$, so it holds when $\gcd(q, 6) = 1$.
- *q even, $d \ge 4$.* Here $m \ge 3$, so both conditions hold.
- *q even, $d = 3$.* By Lemma 2.2 every member P of an Oppermann class has $p_5 = 0$. In characteristic 2 this gives $P' = p_3 t^2 + p_1$, so $\deg P' = 2$ when $p_3 \neq 0$, and the condition "$m \ge 3$ if $\deg P' \le 1$" does not apply.

□

The cases left open by the corollary are the following. For $d = 1$ the interval is too short for [1], but Theorem 1.1(A) and, for odd q, the count $(q-1)/2$ of irreducible $t^2 + bt + s_0$ settle it. For q odd with $d = 2$ and $3 \mid q$ there is no general result. For q even with $d = 2$, Theorem 1.1(B) is complete. For q even with $d = 3$, the q classes with $p_3 = 0$ (prefixes $(p_5, p_4, p_3) = (0, x, 0)$) remain open. In every case, $q < q_0(d)$ is covered only by computation.

6. Questions

1. In characteristic 2, are (A)–(C) the only empty Oppermann sub-intervals for all q and d? By Corollary 5.1, for each $d \ge 4$ only finitely many q can carry further empty sub-intervals. For $d = 3$, the classes with $p_3 = 0$ are not covered by [1], but they are nonempty in the computed range $q \le 16$.
2. In odd characteristic, is every Oppermann sub-interval nonempty for all q and d? Lemma 2.1 turns this into a statement about all short intervals of length $q^{n/2}$. By Corollary 5.1, for each $d \ge 3$ it fails for at most finitely many q. An explicit $q_0(d)$ combined with Theorem 1.2 would settle small d completely.

(3) Outside the characteristic-2 cases above, is there a constant $c > 0$ with $\min_x \pi(x) \geq c\,q^d/2d$ for all large d?

Data and code

- `hayes_oppermann.py`: Theorems 1.2 and 1.3.
- `verify_pointwise_oppermann.py`: brute-force census, $q^{2d} \leq 2^{26}$.
- `compare_bruteforce.py`: class-by-class cross-check.
- `make_tables.py`, `make_figure.py`: Tables 2, 3 and Figure 1.
- `char2_d2_blocks.py`, `char2_d2_theorem.py`: Theorem 1.1(B).
- `swan_parity.py`: Theorem 1.1(C) and Table 1.
- `hayes_fft_fq.py`, `fqlib.py`, `small_q_gf.py`: taken unchanged from [3].

All scripts are in `code/`; their outputs are in `data/`.

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